

Chapter 9

The Instruments of Trade Policy

■ Chapter Organization

Basic Tariff Analysis.

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Effects of a Tariff.

Measuring the Amount of Protection.

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Summary

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■ Chapter Overview

This chapter and the next three focus on international trade policy. Students will have heard in the media various arguments for and against restrictive trade practices. Some of these arguments are sound, and some are clearly not grounded in fact. This chapter provides a framework for analyzing the economic effects of trade policies by describing the tools of trade policy and analyzing their effects on consumers and producers in domestic and foreign countries. Case studies discuss actual episodes of restrictive trade practices. An instructor might try to underscore the relevance of these issues by having students scan newspapers and magazines for other timely examples of protectionism at work.

The analysis presented here takes a partial equilibrium view, focusing on demand and supply in one market, rather than the general equilibrium approach followed in previous chapters. Import demand and export supply curves are derived from domestic and foreign demand and supply curves. There are a number of trade policy instruments analyzed in this chapter using these tools. Some of the important instruments of trade policy include *specific tariffs*, defined as taxes levied as a fixed charge for each unit of a good imported; *ad valorem tariffs*, levied as a fraction of the value of the imported good; *export subsidies*, which are payments given to a firm or industry that ships a good abroad; *import quotas*, which are direct restrictions on the quantity of some good that may be imported; *voluntary export restraints*, which are quotas on trading that are imposed by the exporting country instead of the importing country; and *local content requirements*, which are regulations that require that some specified fraction of a good is produced domestically.

The import supply and export demand analysis assumes a large country tariff, in which the imposition of a tariff drives a wedge between prices in domestic and foreign markets, and increases prices in the country imposing the tariff and lowers the price in the other country by less than the amount of the tariff. This contrasts with most textbook presentations, which make the small country assumption that the domestic internal price equals the world price plus the tariff. The chapter also discusses how the actual protection provided by a tariff may not equal the tariff rate if imported intermediate goods are used in the production of the protected good. The proper measurement, *the effective rate of protection*, is described in the text and calculated for a sample problem.

The analysis of the costs and benefits of trade restrictions require tools of welfare analysis. The text explains the essential tools of consumer and producer surplus. Consumer surplus on each unit sold is defined as the difference between the actual price and the amount that consumers would have been willing to pay for the product. Geometrically, consumer surplus is equal to the area under the demand curve and above the price of the good. Producer surplus is the difference between the minimum amount for which a producer is willing

to sell his product and the price that he actually receives. Geometrically, producer surplus is equal to the area above the supply curve and below the price line. These tools are fundamental to the student's understanding of the implications of trade policies and should be developed carefully.

The costs of a tariff include distortionary efficiency losses in both consumption and production. A tariff provides gains from terms of trade improvement when and if it lowers the foreign export price. Summing the areas in a diagram of internal demand and supply provides a method for analyzing the net loss or gain from a tariff. The gain from a tariff is larger the greater is the decrease in foreign export price from the tariff (as the tariff-imposing country is able to pass some of the costs of the tariff on to foreign exporters). Because large countries will have a larger influence on export prices than small countries, a large country is more likely to gain and, therefore, impose an import tariff. A case study covers the winners and losers of the Trump administration's trade war. The study shows that consumers and the government sector paid higher prices for imported goods. In addition, U.S. agricultural exports fell by 63% following the imposition of tariffs and farm bankruptcies increased by 24% in 2019.

Other instruments of trade policy can be analyzed with this method. An export subsidy operates in exactly the reverse fashion of an import tariff. A text box covers the 17-year long trade dispute over aircraft subsidies and tax rebates. The dispute was finally resolved in 2021. .

An import quota has similar effects as an import tariff upon prices and quantities, but revenues, in the form of quota rents, accrue to the quota license holders, who are often foreign producers. For example, a quota on sugar imported into the United States has greatly increased the fortunes of foreign sugar producers (many of which are owned by American sugar refiners), at a significant cost to American consumers. Estimates place the cost of each job in the American sugar industry "saved" by protection at \$1.75 million, and this number does not include the job losses in the food industry created by higher sugar prices. Total consumer loss is estimated to be over \$3.5 billion in terms of higher prices paid by consumers. Since this loss equates to about \$30 per U.S. household the average consumer is unaware of the costs of protectionism.

Voluntary export restraints are a form of quotas in which import licenses are held by foreign governments. For example, Japan voluntarily limited exports of cars to the United States to forestall any import tariffs on cars from Japan in the wake of the oil price spike of 1979. The net result of these VER's was to raise the price of Japanese cars, with the gains accruing directly to Japanese manufacturers. A similar story is happening now with voluntary export restraints on solar panels exported from China to the European Union.

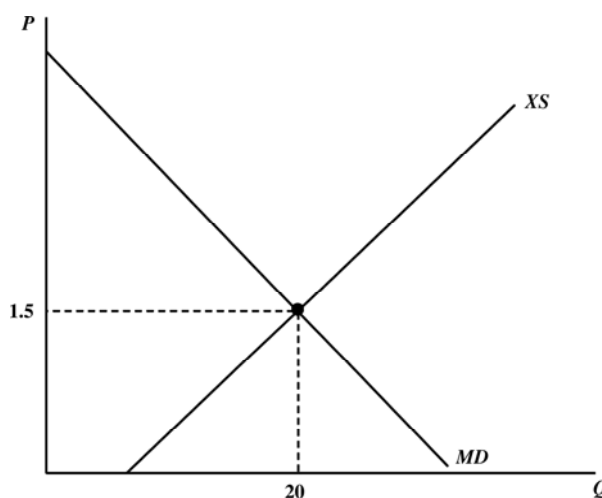
Another trade instrument is to mandate local content requirements. These raise the price of imports as well as domestic goods competing with imports but do not yield either tariff revenue or quota rents. The recent

construction of the new Bay Bridge linking San Francisco and Oakland is used as a case study. Federal funding was available for this project but would have required the state of California to use a much more costly American contractor as opposed to the significantly cheaper Chinese bid. In the end, the bridge was built through local bonds rather than federal funding because of the local content requirement of federal funding.

The Appendix discusses tariffs and import quotas in the presence of a domestic monopoly. Free trade eliminates the monopoly power of a domestic producer, and the monopolist mimics the actions of a firm in a perfectly competitive market, setting output such that marginal cost equals world price. A tariff raises domestic price. The monopolist, still facing a perfectly elastic demand curve, sets output such that marginal cost equals internal price. A monopolist faces a downward-sloping demand curve under a quota. A quota is not equivalent to a tariff in this case. Domestic production is lower and internal price higher when a particular level of imports is obtained through the imposition of a quota rather than a tariff.

■ Answers to Textbook Problems

1. The import demand equation, MD , is found by subtracting the Home supply equation from the Home demand equation. This results in $MD = 80 - 40 \times P$. Without trade, domestic prices and quantities adjust such that import demand is 0. Thus, the price in the absence of trade is 2.
2. a. Foreign's export supply curve, XS , is $XS = -40 + 40 \times P$. In the absence of trade, the price is 1.
- b. When trade occurs, export supply is equal to import demand, $XS = MD$. Thus, using the equations from Problems 1 and 2a, $P = 1.50$, and the volume of trade is 20.



3. a. The new MD curve is $80 - 40 \times (P + t)$ where t is the specific tariff rate, equal to 0.5. (Note: In solving these problems, you should be careful about whether a specific tariff or *ad valorem* tariff is imposed. With an *ad valorem* tariff, the MD equation would be expressed as $MD = 80 - 40 \times (1 + t)P$. The equation for the export supply curve by the foreign country is unchanged.

$$MD = XS$$

$$80 - 40 \times (P + 0.5) = 40P - 40$$

$$80 - 20 - 40P = 40P - 40$$

$$80P = 100$$

$$P^{World} = 1.25$$

$$P^{Home} = P^{World} + t = 1.25 + 0.5 = 1.75$$

$$\text{Trade} = MD = XS = (40 \times 1.25) - 40 = 10$$

$$D^{Home} = 100 - (20 \times 1.75) = 65$$

$$S^{Home} = 20 + (20 \times 1.75) = 55$$

$$D^{Foreign} = 80 - (20 \times 1.25) = 55$$

$$S^{Foreign} = 40 + (20 \times 1.25) = 65$$

- b. and c. The welfare of the Home country is best studied using the combined numerical and graphical solutions presented below in Figure 9-1.

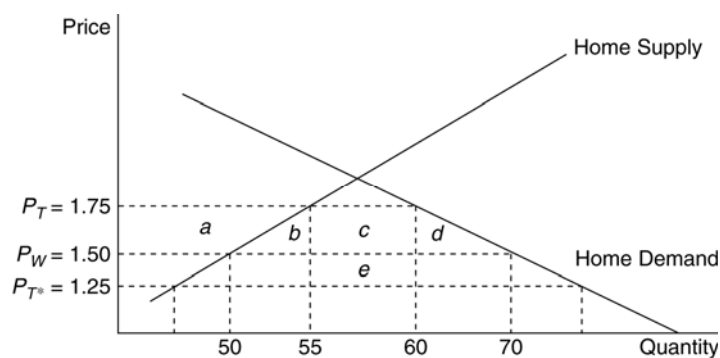


Figure 9-1.

where the areas in the figure are:

$$a. \quad 55(1.75 - 1.50) - 0.5(55 - 50)(1.75 - 1.50) = 13.125$$

$$b. \quad 0.5(55 - 50)(1.75 - 1.50) = 0.625$$

$$c. \quad (65 - 55)(1.75 - 1.50) = 2.50$$

$$d. \quad 0.5(70 - 65)(1.75 - 1.50) = 0.625$$

$$e. \quad (65 - 55)(1.50 - 1.25) = 2.50$$

Consumer surplus change: $-(a + b + c + d) = -16.875$. Producer surplus change: $a = 13.125$. Government revenue change: $c + e = 5$. Efficiency losses $b + d$ are exceeded by terms of trade gain e . (Note: In the calculations for the a , b , and d areas, a figure of 0.5 shows up. This is because we are measuring the area of a triangle, which is one-half of the area of the rectangle defined by the product of the horizontal and vertical sides.)

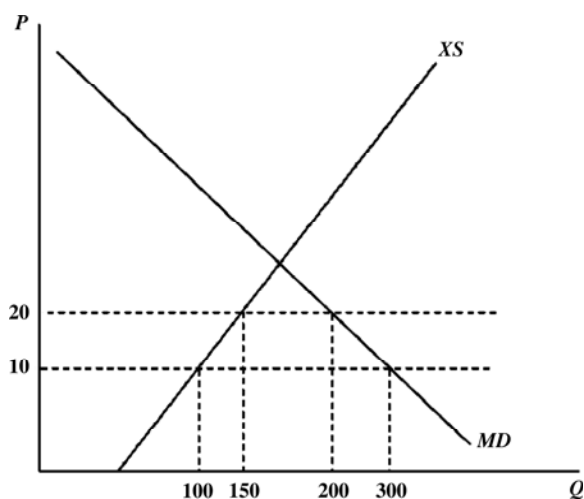
4. Using the same solution methodology as in Problem 3, when the Home country is very small relative to the Foreign country, its effects on the terms of trade are expected to be much smaller. The small country is much more likely to be hurt by its imposition of a tariff. Indeed, this intuition is shown in this problem. The free trade equilibrium is now at the price \$1.09 and the trade volume is now 36.40.

With the imposition of a tariff of 0.5 by Home, the new world price is \$1.045, the internal Home price is \$1.545, Home demand is 69.10 units, Home supply is 50.90, and the volume of trade is 18.20. When Home is relatively small, the effect of a tariff on world price is smaller than when Home is relatively large. When Foreign and Home were closer in size, a tariff of 0.5 by Home lowered world price by 25%, whereas in this case the same tariff lowers world price by about 5%. The internal Home price is now closer to the free trade price plus t than when Home was relatively large. In this case, the government revenues from the tariff equal 9.10, the consumer surplus loss is 33.51, and the producer surplus gain is 21.089. The distortionary losses associated with the tariff (areas $b + d$) sum to 4.14 and the terms of trade gain (e) is 0.819. Clearly, in this small country example, the distortionary losses from the tariff swamp the terms of trade gains. The general lesson is that the smaller the economy, the larger the losses from a tariff because the terms of trade gains are smaller.

5. The effective rate of protection (ERP) is defined as $(V_t - V_w)/V_w$, where V_t is the value added under protection and V_w is the value added under free trade. We define value added as the difference between the price of the final good and the price of components. So, $V_w = \$200 - \$100 = \$100$. With a 50% tariff on bicycles (and a 0% tariff on components), $V_t = (\$200 \times 1.5) - \$100 = \$300 - \$100 = \$200$. Therefore, the $ERP = (200 - 100)/100 = 100\%$.
6. The effective rate of protection takes into consideration the costs of imported intermediate goods. Here, 55% of the cost can be imported, suggesting with no distortion, Home value added would be 45%. A 15% increase in the price of ethanol, though, means Home value added could be as high as 60% (45% Home value added under free trade + 15% increase in the price of ethanol). Effective rate of

protection = $(V_t - V_w)/V_w$, where V_t is the value added in the presence of trade policies, and V_w is the value added without trade distortions. In this case, we have $(60 - 45)/45 = 33\%$ effective rate of protection.

7. We first use Foreign's export supply and Home's import demand curves to determine the new world price. The Foreign supply of exports curve, with a Foreign subsidy of 0.5 per unit, becomes $XS = -40 + 40(1 + 0.5) \times P$. The equilibrium world price is 1.2, and the internal Foreign price is 1.8. The volume of trade is 32. The Foreign demand and supply curves are used to determine the costs and benefits of the subsidy. Construct a diagram similar to that in the text and calculate the area of the various polygons. The government must provide $(1.8 - 1.2) \times 32 = 19.2$ units of output to support the subsidy. Foreign producer surplus rises due to the subsidy by the amount of 15.3 units of output. Foreign consumer surplus falls due to the higher price by 7.5 units of the good. Thus, the net loss to Foreign due to the subsidy is $7.5 + 19.2 - 15.3 = 11.4$ units of output. Home consumers and producers face an internal price of 1.2 as a result of the subsidy. Home consumers surplus rises by $70 \times 0.3 + 0.5(6 \times 0.3) = 21.9$, while Home producer surplus falls by $44 \times 0.3 + 0.5(6 \times 0.3) = 14.1$, for a net gain of 7.8 units of output.
8.
 - a. False, unemployment has more to do with labor market issues and the business cycle than with tariff policy. Empirical estimates suggest that the cost to society of jobs saved through tariffs is exorbitantly high, and tariffs may actually increase unemployment in nonprotected industries.
 - b. False, the opposite is true because tariffs by large countries can actually reduce world prices, which helps offset their effects on consumers.
 - c. This kind of policy might reduce automobile production and Mexico but also would increase the price of automobiles in the United States and would result in the same welfare loss associated with any quota.
9. At a price of \$10 per bag of peanuts, Acirema imports 200 bags of peanuts. A quota limiting the import of peanuts to 50 bags has the following effects:
 - a. Set $MD = 50$ to find the post-quota price: $350 - 15P = 50$. The price of peanuts rises to \$20 per bag.
 - b. The quota rents are $(\$20 - \$10) \times 50 = \$500$.
 - c. The consumption distortion loss is $0.5 \times 100 \text{ bags} \times \$10 \text{ per bag} = \$500$.
 - d. The production distortion loss is $0.5 \times 50 \text{ bags} \times \$10 \text{ per bag} = \$250$.



10. The reason is largely that the benefits of these policies accrue to a small group of people and the costs are spread out over many people. Thus, those who benefit care far more deeply about these policies. These typical political economy problems associated with trade policy are probably even more troublesome in agriculture, where there are long-standing cultural reasons for farmers and farming communities to want to hold onto their way of life, making the interests even more entrenched than they would normally be.
11. It would improve the income distribution within the economy because wages in manufacturing would increase, and real incomes for others in the economy would decrease due to higher prices for manufactured goods. This is true only under the assumption that manufacturing wages are lower than all others in the economy. If they were higher than others in the economy, the tariff policies would worsen the income distribution.